Dielectric Properties of Amorphous Zr-Al-O and Zr-Si-O Thin Films

R. B. van Dover,[1, a)] T. A. Naoi,[1, b)] Hanjong Paik,[1, c)] and M. L. Green[2, d)]
[1)] *Department of Materials Science and Engineering, Cornell University, Ithaca, New York 14853, USA*
[2)] *National Institute of Standards and Technology, Gaithersburg, Maryland, 20899, USA*

(Dated: 22 November 2013)

We have systematically studied the composition dependence of the dielectric properties of $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_{1-x}Si_xO_2$. An essentially linear variation of the static dielectric constant, ϵ_r, was observed as a function of composition, x, for compositions rich in the p-block element, i.e., $x > 0.4$, for both chemical systems. However an abrupt change in ϵ_r is found near $x < 0.35$, associated with the onset of crystallinity in as-deposited films. Breakdown fields do not show a comparable composition dependence. Measurements of the index of refraction at optical frequencies, combined with a simple Clausius-Mossotti interpretation, indicates that low-frequency (ionic) contributions to the polarizability exhibit systematic subtle trends in comparison to values linearly interpolated from the endmembers. These trends are not consistently affected by the presence of crystalline order, but are related to changes associated with heterogeneous local oxygen coordination and bonding.

Keywords: amorphous, dielectrics, oxides, thin films, combinatorial, Zr, Si, Al

I. INTRODUCTION

Amorphous dielectrics comprising mixed transition-metal/p-block oxides are technologically important and generally follow simple trends as a function of composition—but occasionally they exhibit unexpected behavior.,[1],[23] They are widely used in industry as alternatives to the native SiO_2 dielectric used to control charge transport in Si-based field-effect transistors, because early transition metal oxides have a significantly higher dielectric constant (static value of the relative dielectric constant, $\epsilon_s \approx 23$) than SiO_2. Compared to SiO_2, they can therefore achieve a comparable charge density at a specified gate voltage using a layer thick enough to avoid excessive leakage by direct tunneling.[4] Ab-initio studies of amorphous $Zr_{1-x}Si_xO_2$ and $Hf_{1-x}Si_xO_2$ and related materials have highlighted the importance of local structure on dielectric properties.,[5],[6],[78] Polycrystalline and amorphous thin films of these materials exhibit comparable values for ϵ_s; the value for both represents an approximate directional average of the anisotropic value associated with single crystals. As gate dielectrics, amorphous oxides are advantageous over polycrystalline materials since they avoid inhomogeneity in local charge density due to anisotropy that would degrade mobility in the conducting channel and thus the performance of the transistor. Polycrystalline dielectrics are expected to exhibit a lower breakdown field due to the presence of grain boundaries.[?] Thus the composition dependence of crystallization temperature[9] and breakdown field are important to establish. Other properties, such as the presence of fixed charge at the dielectric/semiconductor interface, band offsets with respect to semiconductor and metal gate band energies,[10] and the ability to block diffusion of dopant and metals atoms are also important but are not the subject of the present study.

We used a composition-spread technique for thin film growth to enable composition-dependent trends to be identified with low systematic composition-to-composition process variation very fine composition resolution. The dielectric constant of amorphous films was found to vary roughly linearly with composition, while the dielectric constant of crystalline (transition-metal-rich) films was found to be significantly higher. Leakage currents and breakdown fields were less strongly composition dependent.

II. EXPERIMENTAL METHODS

Compositionally-graded films were prepared using 90°off-axis reactive cosputtering using two 2-inch planar magnetron sputter guns arranged in a facing-targets configuration, as described previously.[11] The 75 mm diameter Si substrates were positioned between the two guns so that species sputtered from each target condense on the substrate simultaneously. The deposition rate decreases rapidly as a function of distance from the target, so a natural composition gradient is obtained in this geometry.[11] The sputter guns were operated at rf power levels chosen to achieve approximately 1:1 cation ratio at the center of the substrate. The films were about 100 nm thick with cations randomly mixed on an atomic scale. Films used for evaluation of crystallinity were deposited on bare Si wafers while those used for evaluation of electrical and optical properties were deposited on wafers pre-coated with sputtered TiN). Prior to film deposition the vacuum chamber was evacuated to a pressure of 10^{-6} Torr

[a)] Electronic mail: vandover@cornell.edu; corresponding author
[b)] Electronic mail: tan42@cornell.edu
[c)] Electronic mail: hp277@cornell.edu
[d)] Electronic mail: martin.green@nist.gov

or lower. Gas flow rates during deposition were 15 std. cm^3/minute (sccm) Ar and 10 sccm O_2, with the turbopump throttled to maintain a pressure of 30 mTorr in the chamber. The substrate was also intentionally biased with 10 W rf power, yielding a dc potential of -75 V that ensures formation of a dense film (avoiding formation of columnar voids during deposition.[12]) The substrate temperature during deposition was maintained below 60℃by attaching the substrate to a heatsink. Under these conditions the deposition rate was roughly 0.1 nm/s,though the rate varies somewhat with position on the sample. The final thickness of the films was on the order of 100 nm.

The composition of the films intentionally varies along the length of the sample. Quantitative composition variation was determined using calibration samples on which a single element was reactively sputtered under the same conditions used for the cosputtered spreads. The thickness of the deposited film, measured as a function of position using a profilometer, was used to calculate the deposition rate. The cation composition of the cosputtered sample is then inferred simply by calculating the ratio of arrival rates of the two cation species. The oxygen stoichiometry in these systems was calculated using charge balance assuming cations are in their highest oxidation state, an assumption that is validated by the observation of low leakage currents in the dielectric. An example of the position dependence of composition is shown in Figure 1. At selected positions the codeposited samples were examined using Rutherford Backscattering Spectroscopy (RBS). The RBS analysis, which has an absolute accuracy estimated to be better than 5%, yielded compositions that were consistent with the results of the rate calibrations, confirming the validity of all of the assumptions inherent in the latter technique. A typical RBS spectrum is shown in the inset to Figure 1.

The thickness of the films is not strictly constant across the length of the sample, since the sum of the deposition rates from the two targets is not constant. The thickness was therefore determined as a function of position by preparing a codeposited sample with a liftoff mask. The mask was then removed and the thickness of the deposited film was determined at 15 positions using a profilometer. These measurements of the dependence of thickness with position were interpolated using a quadratic fitting function.

X-ray diffraction patterns were taken with a Bruker-AXS system[2] equipped with a Cu K_α generator and crossed-wire area (two-dimensional) detector. The resulting data were integrated over the range $47\,\mathrm{deg} < \chi < -133\,\mathrm{deg}$ to yield a single graph of integrated intensity versus 2θ. The crystallites exhibited no preferred orientation. Figure 2 illustrates the reflections associated with monoclinic ZrO_2, observed for Zr-rich compositions that exhibit nanoscale crystalline order. The width of the peaks, corrected for instrument resolution, is a measured of the crystallite size (structural coherence length, L) as inferred using the Debye-Scherrer formula, $L = 0.6\lambda/B cos\theta$ where B is the full width at half maximum. The pattern corresponding to $Zr_{0.75}Si_{0.25}O_2$ are marked with the (hkl) Miller indices of the monoclinic phase; the distinct peak expected for cubic or tetragonal ZrO_2 at approximately 30 deg is clearly not present in the pattern. The pattern corresponding to $Zr_{0.89}Al_{0.11}O_{1.94}$ is less well defined, but is consistent with monoclinic ZrO_2. It is surprising that the monoclinic phase should be present in nanocrystalline $Zr_{0.89}Al_{0.11}O_{1.94}$, both because incorporation of a trivalent ion in ZrO_2 introduces O vacancies that typically stabilize the cubic fluorite phase, as in yttrium-stabilized zirconia, and because grain refinement in unsubstituted ZrO_2 also stabilizes the cubic phase.[?] When an identical $Zr_{1-x}Al_xO_{2-x/2}$ composition spread was annealed at 1000 °C for 20 seconds in N_2, Zr-rich compositions ($x < 0.18$) exhibited a mixture of monoclinic and tetragonal phases while for $0.18 < x < 0.40$ the structure was entirely cubic, a result that is not inconsistent with expected equilibrium behavior. For $x > 0.4$ a mixture of cubic and amorphous material was present. Annealing $Zr_{1-x}Si_xO_2$ at 1000 °C yielded surprisingly similar results, considering that tetravalent Si will not introduce O vacancies to maintain charge balance. Zr-rich compositions ($x < 0.35$) exhibited a mixture of monoclinic and tetragonal phases while for $0.35 < x < 0.80$ the structure is entirely cubic. Apparently introduction of Si, which favors tetrahedral coordination with O (rather than the octahedral coordination of ZrO_2), disrupts the interactions that stabilize the monoclinic structure. Interestingly, no evidence for formation of zircon, $ZrSiO_4$,

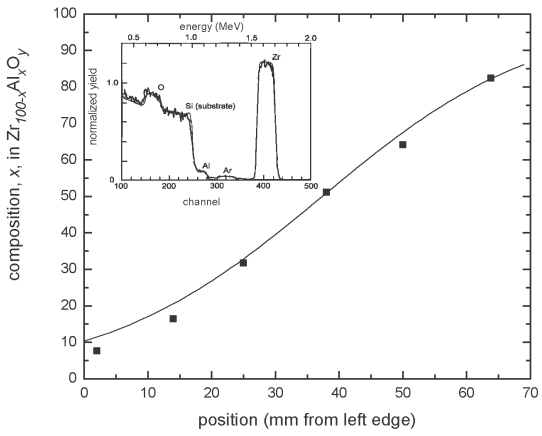


FIG. 1. Composition as a function of position along the 7-cm sample axis. Solid line: compositions calculated using rate calibrations for each species (ZrO_2, Al_2O_3). Squares: Compositions inferred using Rutherford Backscattering Spectroscopy. Inset: a typical RBS spectrum used to determine the composition. A small level of Ar incorporation is typical in sputtered films, and can be seen in the RBS spectrum, though it has a negligible effect on electrical or structural properties.

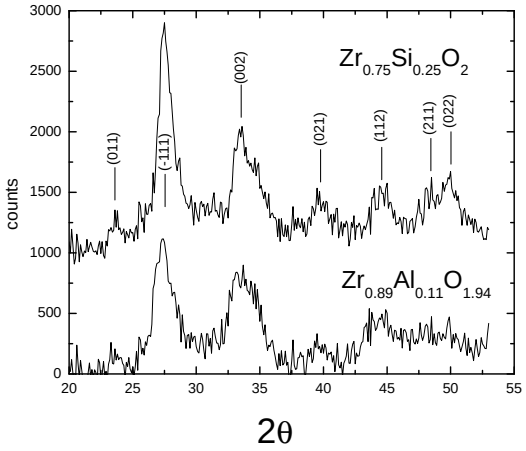


FIG. 2. X-ray diffraction patterns for typical nanocrystalline Zr-rich samples. Top curve: $Zr_{0.75}Si_{0.25}O_2$ film as deposited, offset vertically for clarity, with positions for the strongest monoclinic reflections indicated. Bottom curve: $Zr_{0.89}Al_{0.11}O_{1.94}$, illustrating x-ray peaks that are broader due to the lower degree of crystallinity compared to the Zr-Si-O sample.

is observed in the annealed films.[?]

The index of refraction and thickness of the films were determined using a Sentech SE801 spectroscopic ellipsometer over a wavelength range of 450-800 nm. The optical constants of the substrate layer were determined prior to film deposition and were fixed during the fitting process. A film/substrate bilayer model was used to fit ψ and δ, and the optical constants of the film were modeled using the Cauchy relation $n(\lambda) = n_0 + n_1/\lambda^2 + n_2/\lambda^4$ using a least-squares routine. The reported refractive index n was taken to be n_0, i.e., the infinite wavelength limit far from absorption edge.

The static dielectric constant, ϵ_s was evaluated using parallel-plate capacitors fabricated on oxide films grown on a non-reactive TiN base electrode. Pt counterelectrodes 20 nm thick and 200 μm in diameter were deposited on the films using electron-beam evaporation through a shadow mask. The capacitance of these structures was measured at 10 kHz using an automated probing station and an HP4192A capacitance meter. Some capacitors were measured from 100 Hz to 1 MHz, and were found to exhibit little dispersion over this range. The dissipation factor for all compositions was found to be low ($\tan\delta < 0.05$), indicating high quality films with no anomalous polarization modes.[12] Typically 10 40 capacitors were measured at each composition in the linear composition spread; the median value was chosen as representative. Current-voltage characteristics were also measured on the Pt/dielectric/TiN capacitor structures. A Keithley 617 source/electrometer was used to apply a stepped voltage across the capacitor and to measure the resulting leakage current. Fields from 0 - 10 MV/cm were applied (Pt counterelectrode biased positively), and current densities in the range 10^{-7} – $10^{-3} A/cm^2$ were measured. The breakdown field, E_{br}, was defined as the field at which the current density increased by a factor of 100 or more in a single 0.1 MV/cm step. In cases where the I-V curve was continuous even up to $10^{-3} A/cm^2$ (soft breakdown), the breakdown criterion was chosen to simply be the field at which the current density passed $10^{-4} A/cm^2$. Again, 10 40 capacitors were measured at each composition; because random defects are unavoidable in films fabricated and processed in a non-cleanroom facility we chose the maximum measured E_{br} as the best representative of the intrinsic behavior of the dielectric.

III. RESULTS AND DISCUSSION

A. Dielectric constant and refractive index

The composition dependence of the dielectric constant of $Zr_xAl_{1-x}O_{1.5+x/2}$ is presented in Fig 3. Marks at $x = 0$ and $x = 1.0$ represent measured values for the endmembers deposited under the same conditions ($ZrO_2, \epsilon_s \approx 21$ and $Al_2O_3, \epsilon_s \approx 9$). Also included at $x = 0$ is an estimate of the dielectric constant of cubic ZrO_2 extrapolated from data reported by.[?] The dielectric constant for amorphous (Al-rich) compositions follows a nearly linear interpolation between the values for the amorphous endmenbers. A clear jump in ϵ is seen at approximately $x = 0.35$; extrapolating data for $x < 0.30$ yields a value in good agreement with that inferred for cubic ZrO_2 (although, as discussed above, the films exhibit the monoclinic crystal structure). The transition is associated with the change in crystallinity, as suggested by the top panel of Figure 3, which reports the XRD coherence length (grain size) as a function of composition: the as-deposited films are completely amorphous for $x \gtrsim 0.35$.

The $Zr_{1-x}Si_xO_2$ system exhibits a similar behavior as illustrated in Figure **??**. Although the dielectric constant exhibits a nearly linear trend with composition for amorphous compositions ($0.35 < x < 0.90$) and the trend extrapolates to the measured value for SiO_2 ($\epsilon = 3.9$), for $x = 0$ it extrapolates to $\epsilon = 24$ rather than the measured value for amorphous ZrO_2. As in the Zr-Al-O system the abrupt transition in the trend of ϵ with composition at $x \approx 0.35$ is associated with the change from crystalline to amorphous structure (Figure **??**, top panel). At $x = 0$ the extrapolation of the trend exhibited by crystalline compositions of both Zr-Si-O and Zr-Al-O yields the same value, $\epsilon = 30$, for both, suggesting a common origin. However, we note that the crystal structure observed in these films is not the cubic structure bulk ZrO_2 described by Scott, which also extrapolates to $\epsilon = 30$ for undoped ZrO_2, perhaps coincidentally.

The major trend in dielectric constant as a function of composition in these systems is undoubtedly due to a change in the polarizability from the small values as-

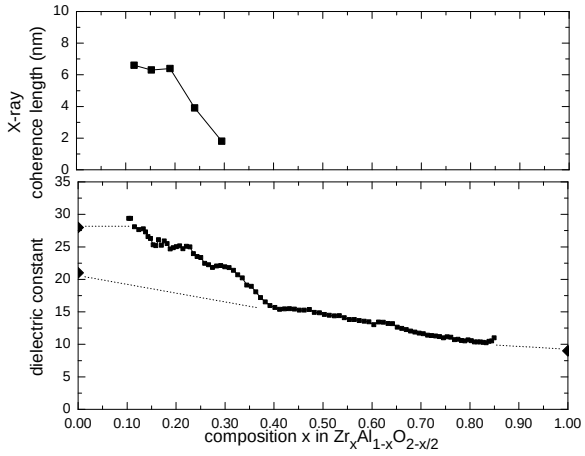


FIG. 3. Bottom: Composition dependence of the dielectric constant of thin film $Zr_xAl_{1-x}O_{1.5+x/2}$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous Al_2O_3) indicated with triangles.Top: composition dependence of the structural coherence length (i.e., essentially grain size) as inferred from X-ray diffraction data using the Debye-Scherrer formula.

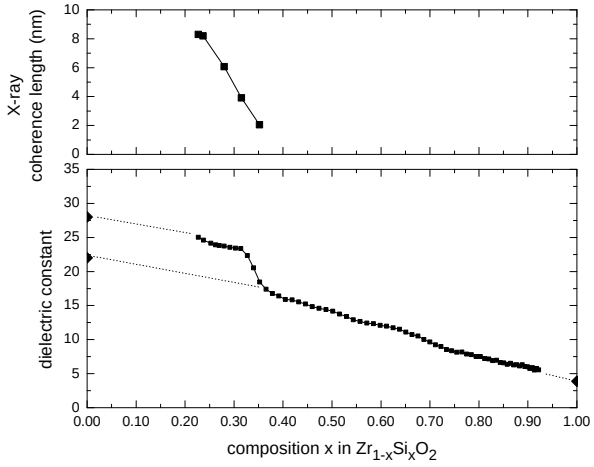


FIG. 4. Bottom: Composition dependence of the dielectric constant of thin film $Zr_xSi_{1-x}O_2$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2) indicated with triangles.Top: composition dependence of the structural coherence length (i.e., essentially grain size) as inferred from X-ray diffraction data using the Debye-Scherrer formula.

sociated with SiO_2 or Al_2O_3 to the large value associated with ZrO_2. But it is not immediately evident what fraction of the low-frequency polarizability can be attributed to ionic motion as opposed to polarization of the electrons with respect to atom cores, nor to what degree the density of the film plays a role. Figure 5

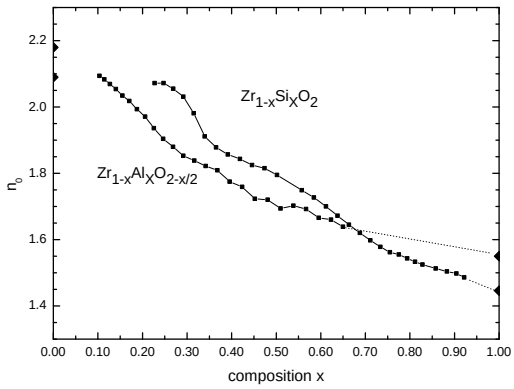


FIG. 5. Bottom: Composition dependence of the refractive index n_0 of thin film $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_xSi_{1-x}O_2$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2 and Al_2O_3) indicated with triangles.

presents the composition dependence of the index of refraction in these systems. The data for $Zr_{1-x}Al_xO_{2-x/2}$ follow a smooth trend that extrapolates almost linearly to accepted values for the endmembers. The data for $Zr_{1-x}Si_xO_2$ exhibits a more dramatic variation, extrapolating linearly to the endmenber value for amorphous SiO_2. The data insufficient to allow unambiguous extrapolation to ZrO_2, though they are consistent with the same value as inferred for $Zr_{1-x}Al_xO_{2-x/2}$. Where the data for Zr-Al-O show only a break in slope associated with the crystalline-amorphous transition, the data for Zr-Si-O exhibit a jump similar to that observed in the static dielectric constant.

B. Microscopic picture: static and electronic polarizability

The high frequency dielectric constant is simply $\epsilon_\infty = n^2$. See supplementary material at [URL will be inserted by AIP] for graphs comparing composition dependence of the static dielectric constant with the ϵ_∞. But comparing ϵ_s with ϵ_∞ does not offer insight into the behavior at the local level because it does not account for depolarization effects and therefore does not account for density variations correctly. Interpreting local polarization behavior in this context does not mean associating a fixed polarizability with specific ions, but is simply a way to compare trends. A baseline expectation might be that the composition dependence of the polarizability interpolates linearly between the values for the endmembers. Deviations from that trend would suggest to unexpected interactions or structural anomalies.

The Clausius-Mossotti equation relates the local polar-

izability to the externally observed dielectric constant:

$$\frac{\epsilon_s - 1}{\epsilon_s + 2} = \frac{4\pi}{3} \sum_m \frac{\alpha_j}{V_m}$$

where α_j is the polarizability due to mechanism $j \in \{electronic, ionic, dipolar \ldots\}$, and V_m is the molar volume. For the materials under consideration only electronic (α_e) and ionic (α_{ion}) mechanisms are present. At optical frequencies, well above any phonon modes, only the electronic mechanism is active, so we can infer α_e/V directly. These values are plotted in Figures 6 and 7, for Zr-Si-O and Zr-Al-O, respectively.

The data for SiO_2-rich Zr-Si-O closely follow a simple linear interpolation between the values for the endmembers; the slight deviations are not thought to be meaningful. For $x < 0.35$ the deviation is significant and is associated with the presence of nanocrystallity. The data for $0.20 < x < 0.30$ are nearly constant, and match the value for unsubstituted ZrO_2. We speculate that introduction of SiO_2 decreases the polarizability and the molar volume at compensating rates, leaving α_e/V constant, and that at $x \approx 0.35$ the transition to a disordered structure leads to a fairly abrupt decrease in the molar volume.

The optical polarizability in the Zr-Al-O system follows a significantly different pattern. Here the trend deviates *below* the linear interpolation of endmember values, suggesting either an increased molar volume or an electronic interaction that lowers the polarizability of the system. The follow a relative smooth curve and extrapolate accurately to the values for the endmembers. In the Zr-Al-O system there is no abrupt transition in α_e associated with the crystalline-amorphous structural transition.

Figures 6 and 7 also show the composition dependence of the static polarizability, α_s/V, to which both electronic and ionic mechanisms contribute. In the $Zr_{1-x}Si_xO_2$ system α_s/V is nearly constant with increasing SiO_2 fraction until at the crystalline/amorphous there is an abrupt decrease, presumably associated with an increase in molar volume. For $x > 0.35$ the trend of $\alpha_s(x)/V$ shows clear negative curvature, leading to a rapid decrease for the most SiO_2-rich compositions. In the $Zr_{1-x}Al_xO_{2-x/2}$ system a less pronounced decrease is associated with the crystalline/amorphous transition, and there is almost no curvature in $\alpha_s(x)/V$ for $x > 0.35$. Fairly abrupt transitions in both the optical and static polarizability are seen around $x = 0.3$ for both chemical systems, clearly associated with the transitions in the dielectric constant. These might be related to changes in molecular density, or could indicate a change in one of the mechanisms for the polarizability.

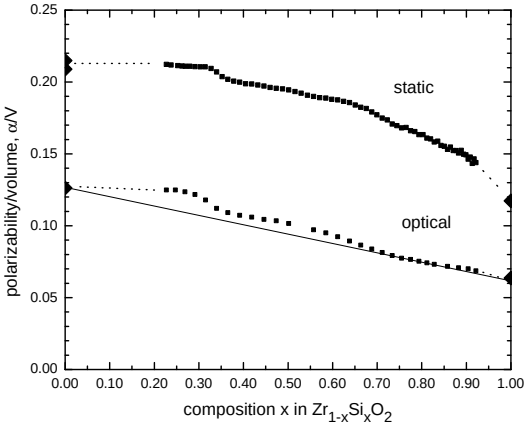


FIG. 6. Composition dependence of the polarizability per unit volume, α/V. Dotted lines are extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2) indicated with triangles. Top curve: polarizability inferred from the static (10 kHz) dielectric constant. Bottom curve: polarizability inferred from the optical index of refraction. The solid line is a simple interpolation between the values for the endmembers.

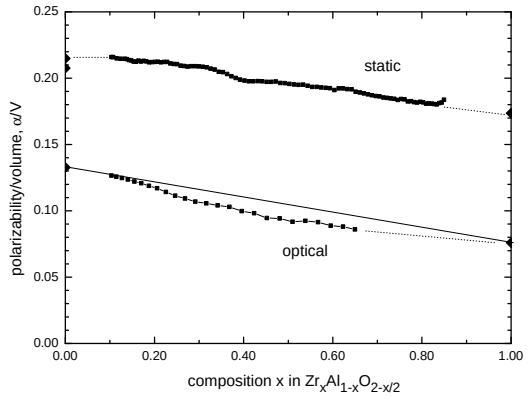


FIG. 7. Composition dependence of the polarizability per unit volume, α/V. Dotted lines are extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous Al_2O_3) indicated with triangles. Top curve: polarizability inferred from the static (10 kHz) dielectric constant. Bottom curve: polarizability inferred from the optical index of refraction. The solid line is a simple interpolation between the values for the endmembers.

C. Microscopic picture; ionic polarizability

The difference between the static and optical dielectric responses is due to the displacement of ions, a mechanism that is active only at phonon frequencies and below. Thus the ionic contribution can be inferred using $\alpha_{ionic}/V = \alpha_s/V - \alpha_e/V$. In Figure 8 we calculate this difference using the data of Figures 6 and 7, in which the optical polarizability was interpolated using a cubic spline before subtracting from the static polarizability. Although the data exhibit small wiggles that are proba-

bly not meaningful, the two systems demonstrate clearly different trends. The data for both compositions spreads can be seen to extrapolate plausibly to standard values inferred for the endmembers. The Zr-Al-O system exhibits a distinct increase in $\alpha_{ionic}/V(x)$ for increasing x up to $x \approx 0.3$, after which α_{ionic}/V is nearly independent of x. The Zr-Si-O system shows almost the opposite trend: a roughly constant value for $\alpha_{ionic}(x)/V$ for $x < 0.7$ followed by a rapid decrease as the SiO_2 is increased further.

At $x = 0.3$, the data $\alpha_{ionic}(x)/V$ in Zr-Al-O do not exhibit a transition comparable to that seen in $\alpha_e(x)/V$; only a modest change of overall slope is observed. If the transition in $\alpha_e(x)V$ were due to a change in molecular volume V, a comparable transition should be observed $\alpha_{ionic}(x)/V$. The lack of such a feature implies that the change is in the polarizability $\alpha_e(x)$ at the crystalline/amorphous transition is due to a reorganization of the electron density in the system. By comparison, the data for Zr-Si-O are consistent the model in which the transition is due to a change in molecular volume V rather than α_{ionic} or α_e.

The composition dependence of the dielectric properties of Zr-Al-O and Zr-Si-O follow trends that are subtly but profoundly different, though this is not readily apparent by observing only the static dielectric constant. The main differences are ascribed to substantial changes in bonding associated with incorporation of Zr_{02} in amorphous SiO_2-rich $Zr_{1-x}Si_xO_2$ and incorporation of Al_2O_3 in ZrO_2-rich $Zr_{1-x}Al_xO_{2-x/2}$ (but not the other way around). We suggest that this reflects, in the first case, the difference in oxygen coordination for Si (typically tetrahedral) compared to that for Zr (typically octahedral). In the second case, it may also reflects the different valence of Al (typically 3+) compared to that of Zr (typically 4+). A more conclusive interpretation of the bonding changes would require data from local probes such as EXAFS, or Raman or infrared spectroscopy, coupled with *ab initio* calculations of average structure in the amorphous materials.

D. Breakdown field

The leakage current and breakdown field of these materials is of significant technological interest, as they affect their suitability for any application in which charge or energy storage is important. Both properties depend on extrinsic mechanism such as details of impurity concentration, local inhomogeneity, film smoothness, electrode composition, etc., so that few general conclusions can be drawn. Figure 9 presents composition dependence of the breakdown field for $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_{1-x}Si_xO_2$. For both materials, the breakdown fields range from *approx*4 MV/cm for the Zr-rich compositions to *approx*6 MV/cm for the Al-rich and Si-rich compositions. The significant feature in this data is that in both systems there is no change in breakdown field associated with the crystalline/amorphous transition at $x \approx 0.35$. This result clearly repudiates the conventional wisdom that the breakdown field of thin film polycrystalline materials must be compromised due to "weak" grain boundaries, as is observed in bulk ceramics. As indicated by the crystallite size plotted in Figures 3 and 4, the film have a very high density of grain boundaries.

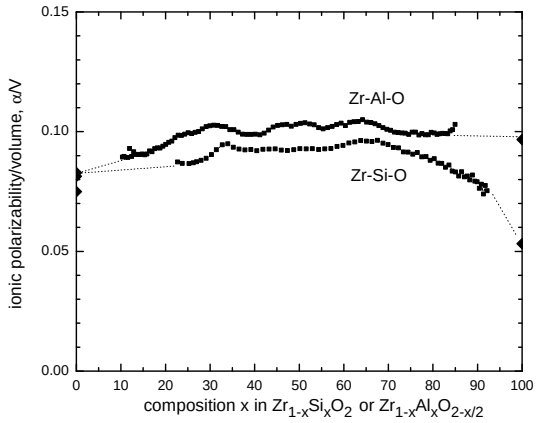


FIG. 8. Composition dependence of the ionic polarizability,, as inferred from the text. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2 and Al_2O_3) indicated with triangles. [add this in text: ZrAlO–smooth trend of enhancement is attributed to perhaps just a smaller molar volume than expected from linear interpolation. ZrSiO apparently a substantial enhancement not likely due to changes in molar volume on the Si rich end; the abrupt change at 35 % might be a change in volume upon crystallization].

IV. CONCLUSION

This work shows that ...

ACKNOWLEDGMENTS

This work was supported by the U.S. Department of Energy under Grant No. DE-FG02-06ER06-15. This work was performed in part at the Cornell NanoScale Facility, a member of the National Nanotechnology Infrastructure Network, which is supported by the National Science Foundation under Grant No. ECS-0335765. Additionally, this work made use of the Cornell Center for Materials Research facilities with support from the National Science Foundation Materials Research Science and Engineering Center program under Grant No. DMR-0520404. Preliminary data was obtained at Lucent Technologies Bell Labs. The authors thank Dr. R. M. Fleming for collecting and interpreting the RBS data.

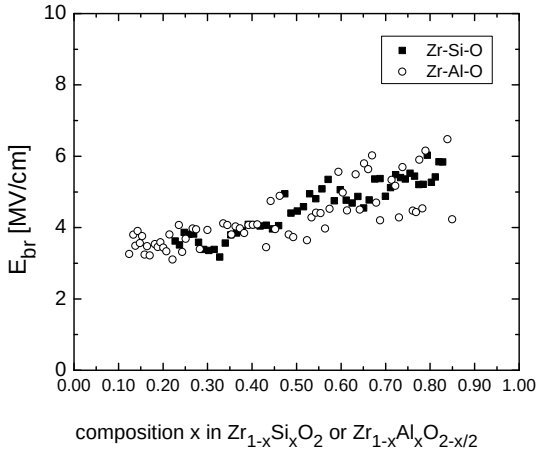


FIG. 9. Composition dependence of the breakdown field in Zr-Si-O and Zr-Al-O. Solid symbols represent Zr-Al-O capacitors and open circles represent Zr-Si-O capacitors. Notice that no change in the trend is associated with the crystalline/amorphous transition at $x \approx 0.35$.

[1] G D Wilk, R M Wallace, and J M Anthony. Hafnium and zirconium silicates for advanced gate dielectrics. *J. Appl. Phys.*, 87(1):484–492, 2000.

[2] G D Wilk, R M Wallace, and J M Anthony. High-k gate dielectrics: Current status and materials properties considerations. *J. Appl. Phys.*, 89(10):5243–5275, May 2001.

[3] G Lucovsky and G B Rayner. Microscopic model for enhanced dielectric constants in low concentration SiO 2 -rich noncrystalline Zr and Hf silicate alloys. *Appl. Phys. Lett.*, 77:2912, 2000.

[4] Angus I Kingon, Jon-paul Maria, and S K Streiffer. Alternative dielectrics to silicon dioxide for memory and logic devices. *Nature*, 406:1032–1038, 2000.

[5] C A Pignedoli, A Curioni, and W Andreoni. Anomalous Behavior of the Dielectric Constant of Hafnium Silicates: A First Principles Study. *Phys. Rev. Lett.*, 98(3):37602, January 2007.

[6] K Nakamura, K Doi, K Fujitani, and A Tachibana. Theoretical study on first-principles dielectric properties of silicate compounds. *Phys. Rev. B (Condensed Matter Mater. Physics)*, 71(4):45331–45332, January 2005.

[7] G-M Rignanese, F Detraux, X Gonze, Angelo Bongiorno, and Alfredo Pasquarello. Dielectric constants of Zr silicates: a first-principles study. *Phys. Rev. Lett.*, 89:117601, 2002.

[8] Lizhi Ouyang and Wai-Yim Ching. Prediction of a high-density phase of SiO2 with a high dielectric constant. *Phys. Status Solidi B*, 242:R64–R66, 2005.

[9] R.B. van Dover, M.L. Green, L.F. Manchanda, L. Schneemeyer, and T. Siegrist. Composition-dependent crystallization of alternative gate dielectrics. *Applied Physics Letters*, 83(7):1459 – 61, 2003/08/18. composition-dependent crystallization;alternative gate dielectrics;amorphous oxides;advanced field-effect transistors;mole fraction;main-group oxide;Zr-Si-O;Zr-Al-O;Hf-Si-O;dielectric constant;silicate systems;aluminates;.

[10] John Robertson. Electronic structure and band offsets of high-dielectric-constant gate oxides. *MRS Bulletin*, 27(3):217 – 221, 2002. Band offsets;High dielectric constant gate oxides;.

[11] R B van Dover, L F Schneemeyer, R M Fleming, and H a Huggins. A high-throughput search for electronic materials-thin-film dielectrics. *Biotechnol. Bioeng.*, 61(4):217–25, 1999.

[12] S.C. Barron, M.M. Noginov, D. Werder, L.F. Schneemeyer, and R.B. van Dover. Dielectric response of tantalum oxide subject to induced ion bombardment during oblique sputter deposition. *Journal of Applied Physics*, 106(10):104110 (9 pp.) –, 2009/11/15. insulating tantalum oxide thin films;ion bombardment;olique sputter deposition;low frequency dielectric constant;low frequency loss tangent;Maxwell-Wagner type polarization;columnar voids;void structure;dielectric films;capacitive measurement;Ar¡sup¿+¡/sup¿ flux;on-axis reactive sputtering;Ta¡sub¿2¡/sub¿O¡sub¿5¡/sub¿;.

[13] T A Naoi, S C Barron, M Noginov, and R B van Dover. Dielectric enhancement in amorphous TaxGe1-xOy thin films. *Appl. Phys. Lett.*, 101:92901, 2012.